FACULTY OF ENGINEERING AND BASIC SCIENCES
ACADEMIC PROGRAM: DATA ENGINEERING AND ARTIFICIAL INTELLIGENCE

COURSE: ETL (G01)
ETL-Project: Colombia & Sustainable Development Goals
First Delivery

1. Objective

Design and implement the first stage of a production-grade ETL pipeline using Colombian open government datasets related to one Sustainable Development Goal (ODS).

In this first delivery, students must design and implement the foundational components of an ETL pipeline, including data acquisition, transformation, dimensional modeling, and database loading.

The focus of this delivery is on data architecture and structured data integration — not only exploratory analysis.

2. Scope of Work

2.1 Business Understanding & Requirements

- Define a business scenario: Select one SDG related to Colombia.
- Clearly define:
  - Business objective
  - Analytical questions to answer
  - Expected KPIs
- Define evaluation criteria for success.

2.2 Data Source Selection

- Select one main open government dataset with:
  - Minimum 50,000 rows
  - Minimum 10 attributes
- Justify:
  - Relevance
  - Structure
  - Data quality
  - Potential analytical value

You can use a CSV dataset as raw input.

2.3 EDA, Data Profiling & Quality Assessment

Perform an initial profiling including:

- Null value analysis
- Duplicate detection

- Data type validation
- Outlier detection
- Inconsistent formatting detection

Document all quality issues identified.

2.4 Define the Grain (MANDATORY)

Before designing the data model:
- Clearly define the grain of your fact table.
- Example: One row per product sale transaction.

No schema design will be accepted without an explicitly defined grain.

2.5 Dimensional Data Model

Design a dimensional model including:
- At least:
  - 1 fact table
  - 2 dimension tables
- Surrogate keys required.
- Explicit primary and foreign keys.
- Clearly distinguish:
  - Measures
  - Dimensions

Provide:
- Star schema diagram
- Explanation of modeling decisions

2.6 ETL Process Implementation

Develop a Python-based ETL pipeline that performs:

Extract
- Read raw CSV

Transform
- Data cleaning
- Type casting
- Derived columns
- Standardization
- Surrogate key generation
- Basic validation rules

Load
- Load data into relational database
- Ensure referential integrity

2.7 Database Layer

Use a relational database (PostgreSQL, MySQL, etc.).

2.8 KPIs and Visualizations

All reports must be generated **from the Data Warehouse**, not from the CSV.
Build meaningful charts, dashboards and/or reports.

3. Technologies

You must use:

- Python
- Jupyter Notebook
- SQL
- A Data Warehouse of your choice (PostgreSQL, MySQL, BigQuery, Snowflake, etc.)
- BI Tool (Power BI, Tableau, etc.)

4. Proposed System Architecture

The following diagram illustrates the high-level architecture expected for this project. It represents the complete ETL data flow, from raw data ingestion to analytical consumption.

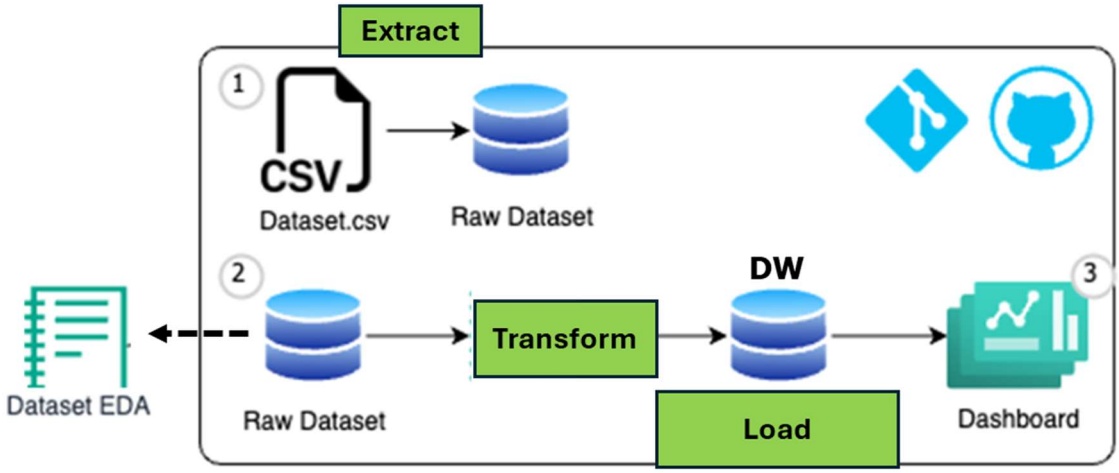

Figure 1. Proposed System Architecture

5. GitHub Deliverables

Your repository should include:

- **ETL Code** (code for Extract, Transform, Load).
- **Star Schema Diagram** (image + explanation).
- **Visualizations** (e.g., exported as screenshots).
- **README.md** explaining your project, setup instructions, and key decisions.

- **.gitignore** file.

README Requirements

Your README must explain:

- Project objective
- Star Schema design decisions
- Grain definition
- ETL logic
- Data quality assumptions
- How to run the project
- Example outputs

6. Evaluation